DO NOW — RECALL

# Data structure selection challenge

Name: _____ Class: _____ Date: _____

**1. What is an accumulator?**

**2. What is a collision?**

**3. What is a head pointer?**

**4. Which structure is used for Undo and why?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

A variable storing a running total.

### 2. Model answer

Two keys map to the same initial bucket.

### 3. Model answer

A reference to the first node.

### 4. Model answer

A stack: the most recent operation must be undone first.

### Vocabulary — know these terms

LIFO — Last In, First Out.

FIFO — First In, First Out.

lookup — Retrieve a value using a key.

balanced BST — Tree kept at relatively small height.

trade-off — Benefit balanced against a drawback.

NEW LEARNING

# Read, annotate and explain

### Identify the operation

The print job that arrived first must leave first.

### Match the structure

Enqueue adds jobs; dequeue removes the oldest one. This is a queue.

### Justify and qualify

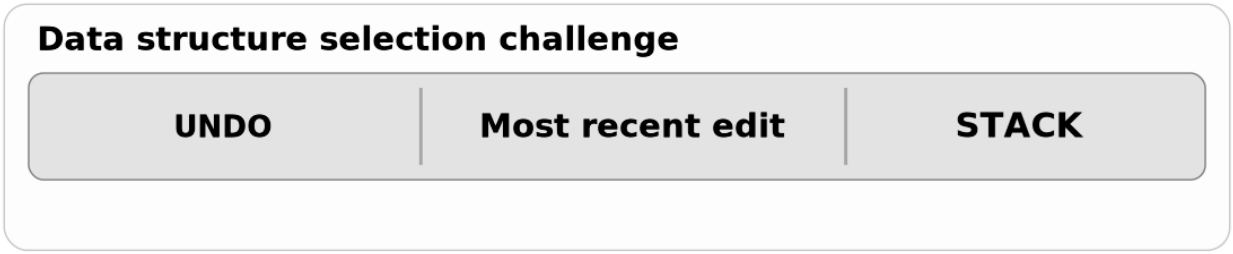
A stack gives LIFO, which is the wrong order. Other systems may also need priority queues if priority overrides arrival order.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Data structure selection challenge: UNDO, Most recent edit, STACK

### New Learning




Data structure selection challenge: UNDO, Most recent edit, STACK; PRINTER, Oldest job first, QUEUE

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Data structure selection challenge: UNDO, Most recent edit, STACK; PRINTER, Oldest job first, QUEUE; LOOKUP, Exact key required, HASH TABLE

I DO — WORKED EXAMPLE

# Follow the worked method

**A browser Back button must return to the most recently visited page. Choose a structure.**

### Predict

The last visited page is the first one that must be removed.

### Explain

Use a stack: push visits and pop the most recent page.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A browser Back button must return to the most recently visited page. Choose a structure.

The last visited page is the first one that must be removed.




A browser Back button must return to the most recently visited page. Choose a structure.: VISIT, Home then News, PUSH

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The last visited page is the first one that must be removed.




A browser Back button must return to the most recently visited page. Choose a structure.: VISIT, Home then News, PUSH; LATEST PAGE, Sport, TOP

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A browser Back button must return to the most recently visited page. Choose a structure.

The last visited page is the first one that must be removed.




A browser Back button must return to the most recently visited page. Choose a structure.: VISIT, Home then News, PUSH; LATEST PAGE, Sport, TOP; BACK, Remove Sport, STACK

WE DO — GUIDED PRACTICE

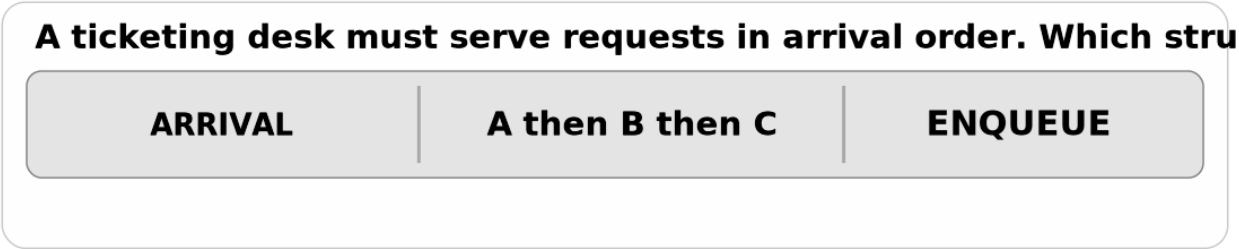
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A ticketing desk must serve requests in arrival order. Which structure?

The first request A must be the first one handled.




A ticketing desk must serve requests in arrival order. Which structure?: ARRIVAL, A then B then C, ENQUEUE

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict The first request A must be the first one handled.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Select a structure for processing jobs in arrival order.

### Question 2

Choose stack or hash table to retrieve customer records by unique ID. Justify.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Compare a balanced BST and hash table for lookup and sorted output.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A hash table is always O(1) even in the worst case.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

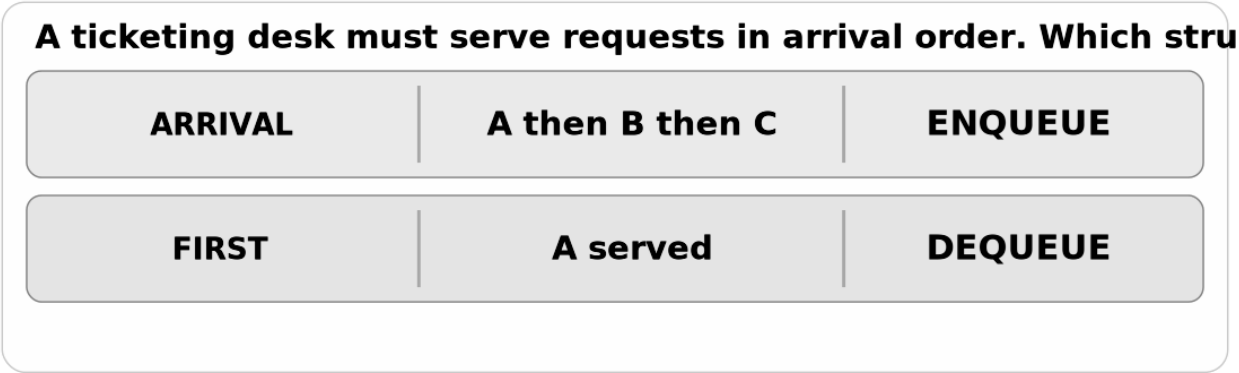
Next step: revisit the model or a relevant revision-guide section if needed.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A ticketing desk must serve requests in arrival order. Which structure?

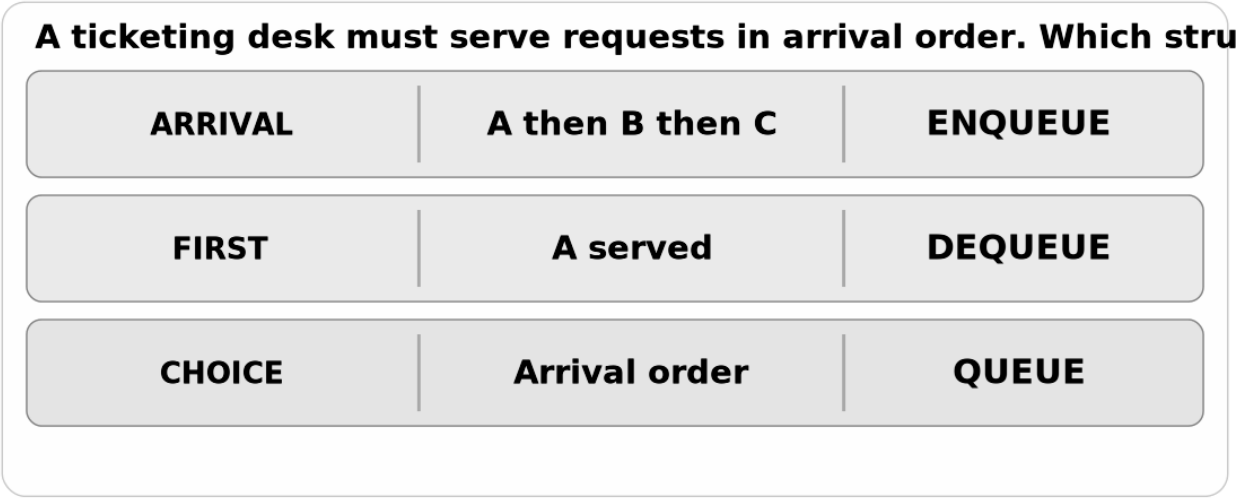
The first request A must be the first one handled.




A ticketing desk must serve requests in arrival order. Which structure?: ARRIVAL, A then B then C, ENQUEUE; FIRST, A served, DEQUEUE

### A ticketing desk must serve requests in arrival order. Which structure?

The first request A must be the first one handled.




A ticketing desk must serve requests in arrival order. Which structure?: ARRIVAL, A then B then C, ENQUEUE; FIRST, A served, DEQUEUE; CHOICE, Arrival order, QUEUE

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Explain

Use a queue, because FIFO removes the oldest item.

IMPROVE: Explain one correction to your own reasoning.